DSC 140B - Quiz 02

Name: 　　　　　　 PID:

About the quizzes:

- Quizzes in DSC 140B are *optional* and graded pass/fail.
- A score of 70% or higher earns a "pass" and 1.5 credits toward your final grade.
- If you don't pass, no credits are earned, but it doesn't hurt your grade.
- You have 30 minutes to complete the quiz.
- At least one of the questions below will be on an exam (probably with slight changes, such as different numbers).
- Unfortunately, we can't answer clarifying questions during the quiz. If you think a question has a bug or is unclear, please let us known in a private post on Campuswire after the quiz, and we'll take it into account when grading.

Problem 1.

Let $\vec{x} = (3, -1, 4, 2, -5)^T \in \mathbb{R}^5$. What is $\vec{x} \cdot \hat{e}^{(4)}$?

Problem 2.

True or False: The following three vectors form an orthonormal basis for $\mathbb{R}^3$.

$$\vec{v}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}, \quad \vec{v}_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

○ True

○ False

Problem 3.

True or False: $\vec{f}(\vec{x}) = (x_1 + 4,\, x_2 - 1)^T$ is a linear transformation.

○ True

○ False

Problem 4.

Suppose $\vec{f}$ is a linear transformation with:

$$\vec{f}(\hat{e}^{(1)}) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \quad \vec{f}(\hat{e}^{(2)}) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Let $\vec{x} = (3, 2)^T$. Compute $\vec{f}(\vec{x})$.

- ○ $(8, -3)^T$
- ○ $(3, 2)^T$
- ○ $(5, 5)^T$
- ○ $(8, 3)^T$

Problem 5.

Suppose $\vec{f}$ is a linear transformation with:

$$\vec{f}(\hat{e}^{(1)}) = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad \vec{f}(\hat{e}^{(2)}) = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$$

Which matrix A represents $\vec{f}$ with respect to the standard basis?

- ○ $\begin{pmatrix} 4 & 1 \\ -2 & 5 \end{pmatrix}$
- ○ $\begin{pmatrix} -2 & 4 \\ 5 & 1 \end{pmatrix}$
- ○ $\begin{pmatrix} 4 & -2 \\ 1 & 5 \end{pmatrix}$
- ○ $\begin{pmatrix} 1 & 5 \\ 4 & -2 \end{pmatrix}$

Problem 6.

Consider the linear transformation $\vec{f} : \mathbb{R}^2 \to \mathbb{R}^2$ that rotates vectors by $180°$.

What is the matrix A that represents $\vec{f}$ with respect to the standard basis?

- ○ $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
- ○ $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
- ○ $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$
- ○ $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

Problem 7.

Let $\hat{u}^{(1)} = \frac{1}{\sqrt{2}}(1,1)^T$ and $\hat{u}^{(2)} = \frac{1}{\sqrt{2}}(1,-1)^T$ form an orthonormal basis $\mathcal{U}$ for $\mathbb{R}^2$.

Given $\vec{x} = (3,1)^T$, what is $[\vec{x}]_{\mathcal{U}}$? That is, what are the coordinates of $\vec{x}$ in the basis $\mathcal{U}$?

- ◯ $(4,2)^T$
- ◯ $(2,1)^T$
- ◯ $(2\sqrt{2},\sqrt{2})^T$
- ◯ $(\sqrt{2},2\sqrt{2})^T$

Problem 8.

Let $\hat{u}^{(1)} = \frac{1}{\sqrt{2}}(1,1)^T$ and $\hat{u}^{(2)} = \frac{1}{\sqrt{2}}(1,-1)^T$ form an orthonormal basis $\mathcal{U}$ for $\mathbb{R}^2$.

Suppose $[\vec{x}]_{\mathcal{U}} = (2\sqrt{2},3\sqrt{2})^T$. That is, the coordinates of $\vec{x}$ in the basis $\mathcal{U}$ are $(2\sqrt{2},3\sqrt{2})^T$.

What is $\vec{x}$ in the standard basis?

- ◯ $(5,-1)^T$
- ◯ $(-1,5)^T$
- ◯ $(5\sqrt{2},-\sqrt{2})^T$
- ◯ $(2\sqrt{2},3\sqrt{2})^T$

Problem 9.

Let $\hat{u}^{(1)}$ and $\hat{u}^{(2)}$ be orthogonal unit vectors. We do not know their components, but we do know that they are orthonormal.

What is $(3\hat{u}^{(1)} + 2\hat{u}^{(2)}) \cdot (5\hat{u}^{(1)} + \hat{u}^{(2)})$?

- ◯ 17
- ◯ 15
- ◯ 30
- ◯ 11

Problem 10.

Suppose $\vec{v} = (3,4)^T$.

Find a unit vector $\hat{u}$ such that $|\hat{u} \cdot \vec{v}|$ is maximized.

- ◯ $(3/\sqrt{7},4/\sqrt{7})^T$
- ◯ $(1/\sqrt{2},1/\sqrt{2})^T$
- ◯ $(3/5,4/5)^T$
- ◯ $(4/5,3/5)^T$
- ◯ $(3,4)^T$
- ◯ $(-4/5,3/5)^T$